Carla Trapp

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SUMMARY

ETH Zurich MSc Chemistry researcher with theoretical grounding in quantum chemistry and computational chemistry, drawn to hard problems and the challenge of finding algorithmic solutions, with research experience spanning machine learning, high-performance computing, and neuromorphic computing.

EDUCATION

ETH Zurich — Zurich, Switzerland
MSc Chemistry – Focus: Computational & Theoretical Chemistry (Current grade: 5.4/6.0) — *09/2024 – 08/2026*

Imperial College London — London, United Kingdom
Exchange Semester (Selected grades: A, A, A) — *09/2024 – 01/2025*

ETH Zurich — Zurich, Switzerland
BSc Chemistry (Grade: 5.12/6.0) — *09/2021 – 08/2024*

Schloss Hansenberg Boarding School — Geisenheim, Germany
Abitur – State boarding school for gifted & high-performing pupils (Grade: 1.0/1.0) — *08/2018 – 06/2021*

RESEARCH AND WORK EXPERIENCE

Institute of Molecular Physical Science (ETH Zurich) — Zurich, Switzerland
MSc Thesis — Supervisors: Prof. Dr. Markus Reiher, Dr. Can Liao — *03/2026 – 08/2026*

- Developing a low-scaling VSCF algorithm in Colibri with arbitrary-order numerical differentiation, parallelized via MPI/OpenMP for HPC execution.

Institute of Neuroinformatics (UZH & ETH Zurich) — Zurich, Switzerland
Research Project — Supervisors: Prof. Dr. Guillén Gosálbez, Prof. Dr. Indiveri — *06/2025 – 12/2025*

- Developed a Python framework mapping NP-hard combinatorial problems to Spiking Neural Networks (SNNs) on mixed-signal neuromorphic hardware, applied to chemical synthesis optimization.

Institute for Chemical and Bioengineering (ETH Zurich) — Zurich, Switzerland
Research Project — Supervisors: Prof. Dr. Andrew de Mello, Dr. Nathan Khosla — *02/2024 – 07/2024*

- Built a computer vision pipeline for automated analysis of diagnostic assay images, achieving 99% well detection in low-contrast conditions.
- Selected as sole undergraduate speaker for an international seminar.

Institute of Energy and Process Engineering (ETH Zurich) — Zurich, Switzerland
Student Research Assistant — Supervisors: Prof. Dr. André Bardow, Dr. Lisa Rueben — *03/2023 – 02/2024*

- Modeled electrolyte permittivities using ePC-SAFT thermodynamic equations of state in Python.

Women In Natural Sciences (ETH Zurich) — Zurich, Switzerland
Board Member & Webmaster (Backend & Frontend) — *03/2024 – Present*

- Maintaining and developing website infrastructure (Apache, MySQL) on a Linux virtual machine.

SELECTED AWARDS

- **Valedictorian in Chemistry** – Awarded by German Chemical Society (06/2021)
- **Honorary Pin for Outstanding Performance** – Schloss Hansenberg Boarding School (02/2021)
- **Top 5 Team in Germany** – Bolyai International Mathematics Competition (01/2020)

SKILLS AND INTERESTS

- **Programming Languages:** Python, C++, R, Q#, MATLAB, MySQL
- **Software & Frameworks:** Keras, SciKit-Learn, OpenCV, PIL, RDKit, Optuna, Git, Qiskit, Seaborn
- **Relevant Coursework:** Quantum Chemistry, Molecular Quantum Mechanics, Introduction to Machine Learning (for Computer Scientists), Chemometrics and Machine Learning (for Chemical Engineers), Digital Chemistry, Algorithms and Programming for Chemistry
- **Certifications:** IBM Machine Learning Professional, Basics of Quantum Information, Practical Intro to Quantum-Safe Cryptography
- **Languages:** German (Native), English (C1), French (B1), Latin (Latinum)
- **Soft Skills:** Analytical Thinking, Curiosity, Dedication
- **Hobbies:** Solving cryptic Sudokus, Dancing, Reading